

TEAM REFLECTION

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Project Title: Modeling Energy Loss in a Shock Absorber

Course: Dynamic System Modeling (702MH0C023)

B.Tech Mechatronics, Sem IV | NMIMS MPSTME Mumbai

Category: D (Mechanical Systems) | Complexity: Medium

Date: [20/04/2026]

Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

- The main challenge was accurately modeling the damping and spring stiffness interaction to capture energy dissipation over compression and extension cycles.
- The difficulty came from coupling the nonlinear damping term (proportional to velocity) with varying spring compression rates.
- We resolved this by analyzing simulation outputs from PyBullet that tracked spring force, displacement, and velocity over time, and then applying numerical integration to compute energy loss per cycle using the area under the force–displacement curve.

Q2: Which Course Outcome (CO1–CO5) was most challenging for your team? Why?

- CO4 – Implementing and interpreting numerical simulations for dynamic systems was the hardest.
- This is because integrating the physical model with PyBullet required understanding URDF dynamics, correctly defining prismatic joint behavior, and controlling the alternating forces (+1500 N, -1250 N) to simulate realistic oscillation.
- Achieving stable simulation cycles without divergence forced the team to fine-tune simulation parameters like time step and pause duration.

Q3: What would you do differently if you started the project over?

- Start by building the simplified analytical model first, then validate it with simulation instead of the other way around.
- Include experimental damping coefficients earlier to better match realistic energy losses.
- Automate data extraction from simulation logs for faster plotting and energy computations across multiple cycles.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

- How dynamic systems behave under non-ideal, time-dependent forces — something static textbook problems ignore.
- The visualization of spring compression and extension phases (red/green/yellow in the simulation) gave insight into transient response, phase lag, and damping behavior.
- We also learned to handle numerical instabilities and real-time simulation control in a physics engine.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

- One member handled URDF and PyBullet setup, another conducted data collection and plotting, and the third derived the mathematical energy loss model ($E = \int F \cdot dx$).
- All members discussed model validation and reviewed derived equations to ensure everyone understood how force, velocity, and displacement contributed to energy dissipation.
- Team review sessions helped ensure shared understanding before compiling final results.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

- Used AI-assisted coding (ChatGPT) to debug Python-PyBullet force control loops and improve plotting scripts.
- Used AI for documentation summarization and code comments generation.
- Did NOT use AI for deriving the physical equations or performing numerical integration — those were handled by the team to maintain conceptual rigor and learning integrity.

Summary

Insight:

The simulation demonstrated that energy loss per compression-extension cycle increases with higher applied force amplitude and shorter pause durations, effectively showing how damping stabilizes oscillations in a real shock absorber.